

MAHESH KUMAR

DevOps Architect | Cloud Infrastructure | Platform Engineering

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PROFESSIONAL SUMMARY

Results-driven DevOps Architect with 7 years of experience designing, automating, and scaling cloud-native infrastructure across AWS and Azure. Proven track record of architecting multi-cloud Kubernetes platforms (EKS, AKS), building reusable IaC frameworks with Terraform, and establishing enterprise-grade observability stacks. Experienced in leading cross-functional teams, mentoring engineers, and driving platform engineering culture. Adept at reducing operational toil, improving deployment velocity, and aligning infrastructure strategy with business outcomes.

CAREER HIGHLIGHTS

7 Years DevOps & cloud infrastructure experience	130 Repos Migrated Bitbucket to GitHub with zero CI/CD downtime
5 Major Clients Delivered end-to-end DevOps transformations	100+ Microservices orchestrated on Nomad/Consul cluster
Zero Downtime Led VM to AKS, Nomad to EKS & SCM migrations	DevSecOps SAST, secrets detection & security-gated pipelines

TECHNICAL SKILLS

▪ Kubernetes (EKS, AKS)	▪ AWS	▪ Azure
▪ Terraform (IaC)	▪ Docker	▪ Ansible
▪ Jenkins & Shared Libraries	▪ Azure DevOps	▪ GitHub / Bitbucket
▪ Python Automation	▪ Nomad & Consul	▪ CI/CD Pipeline Design
▪ Prometheus & Alertmanager	▪ Grafana Dashboards	▪ Datadog APM
▪ Nginx & HAProxy	▪ Linux Administration	▪ DevSecOps (SAST, Secrets Scan)

WORK EXPERIENCE

Lead DevOps Engineer / DevOps Architect | **OpsTree Solutions**

Mar 2019 – Present (7 Years)

Client: **Apple Hyderabad** — On-Premises & AWS Hybrid Infrastructure

- Architected and managed a Nomad + Consul cluster orchestrating **100+ microservices** on-premises, maintaining high availability across production workloads.
- Designed and led migration of services from Nomad to **Amazon EKS**, provisioning all AWS resources via Terraform — enabling auto-scaling and cloud-native resilience.
- Developed a Python utility to fully automate SSL/TLS certificate renewal in the secret management system, eliminating manual certificate rotation risk.
- Integrated Grafana for real-time certificate visibility; optimized on-call workflows by refining alert thresholds — reducing alert noise by ~40% and improving MTTR.
- Authored alert runbooks (one-pagers) adopted by the entire on-call team, improving incident response consistency.
- Managed Nginx and HAProxy for load balancing and traffic routing to containerized services; integrated Java apps with **Datadog** for APM and service health monitoring.
- Automated recurring server administration tasks using Ansible, reducing manual operational overhead.

Client: **Pine Labs** — CI/CD Modernization, SCM Migration & DevSecOps

- Led end-to-end migration of **130 repositories** from Bitbucket to GitHub with **zero downtime** and no CI/CD disruption — preserving full branch history, PR workflows, and pipeline continuity throughout the transition.
- Architected Jenkins infrastructure on AWS from scratch using **Terraform** — provisioning master and dynamic EC2-based slave nodes, enabling on-demand build capacity and significantly reducing idle compute costs.
- Automated Jenkins setup and configuration using **Ansible** — ensuring consistent, repeatable, and version-controlled Jenkins deployments across environments.
- Redesigned all CI/CD pipelines using **Jenkins Shared Libraries** — eliminating duplicated pipeline code across 130 repos, enforcing standardization, and cutting pipeline maintenance effort by ~70%.
- Drove **pipeline optimization** initiatives including parallel stage execution, caching strategies, and build artifact reuse — reducing average build times and accelerating delivery cycles.
- Introduced **DevSecOps practices** into the CI/CD pipeline — integrating SAST, dependency vulnerability scanning, and secrets detection as mandatory pipeline gates, embedding security into every build.

Central Platform Team — Internal Tooling & Reusable Frameworks

- Architected a library of reusable **Terraform modules** for AWS resource provisioning, adopted across **multiple client projects** — cutting provisioning time by ~60%.
- Established and deployed the Prometheus observability stack via Ansible for the Odmont project, providing unified metrics collection across services.
- Designed custom Grafana dashboards for databases and middleware, giving engineering teams real-time visibility into system health.
- Mentored junior engineers on Terraform best practices, Ansible automation, and CI/CD pipeline design — contributing to a **stronger internal DevOps capability** within the team.
- Built a generic, reusable CI/CD pipeline framework for Java-based applications, standardizing the delivery process across teams.

Client: Publicis Sapient — Azure Cloud Modernization

- Automated build and release cycles using Azure DevOps pipelines, improving deployment frequency and reducing release risk.
- Managed configuration of multi-node **Cassandra, Kafka, and SonarQube** clusters using Ansible roles and playbooks, replacing manual server setup entirely.
- Established full-stack observability using Prometheus, Grafana, Telegraf, and Alertmanager, covering Linux machines, Kafka, Cassandra, Redis, and Zookeeper.
- Led migration of the entire application stack from Azure App Services to **AKS**; provisioned all infrastructure using Terraform on Azure, achieving improved scalability and cost efficiency.

Client: Connect Minda — Azure Kubernetes & Monitoring Architecture

- Designed and led migration from Virtual Machines to **Azure Kubernetes Service (AKS)**, improving application scalability and operational manageability.
- Architected and implemented end-to-end CI/CD pipelines in Azure DevOps for automated AKS deployments, enabling seamless zero-touch delivery.
- Designed and executed a full Prometheus monitoring stack on AKS, covering all deployed services and infrastructure components.
- Configured Prometheus alerting integrated with Microsoft Teams and Slack for real-time incident response and notifications.
- Created custom Grafana dashboards visualizing key performance metrics, application health, and cluster resource utilization for stakeholder reporting.

NOTABLE PROJECTS & CONTRIBUTIONS

- **Nomad to EKS Migration (Apple):** Architected the full migration strategy and execution plan for moving 100+ production microservices from on-prem Nomad to AWS EKS — improving resilience, auto-scaling, and cloud cost visibility.
- **SCM Migration & CI/CD Modernization (Pine Labs):** Designed and executed the zero-downtime migration of 130 repositories from Bitbucket to GitHub; rebuilt Jenkins infrastructure on AWS using Terraform + Ansible; standardized all pipelines via Shared Libraries and introduced DevSecOps security gates.
- **Terraform Module Library:** Built and maintained a centralized library of reusable Terraform modules used across 5+ client engagements at OpsTree, reducing infrastructure provisioning time by ~60%.
- **Enterprise Observability Framework:** Designed and deployed unified Prometheus + Grafana + Alertmanager stacks across 4 enterprise clients, enabling standardized monitoring and on-call practices.
- **Certificate Automation Utility (Python):** Built a Python-based automation tool for certificate lifecycle management integrated with a secrets manager, eliminating manual rotation risk across production services.

- **Open Source & GitHub (github.com/k3mahesh):** Actively shares DevOps scripts, automation utilities, and infrastructure templates on GitHub — demonstrating continuous learning and community contribution.

LEADERSHIP & MENTORING

- Acted as the go-to DevOps architect for client engagements, making infrastructure design decisions and translating business requirements into scalable platform solutions.
- Mentored junior and mid-level DevOps engineers on Kubernetes, Terraform IaC, CI/CD design, and observability best practices.
- Created internal documentation, runbooks, and alert one-pagers adopted across on-call teams — reducing incident resolution time and onboarding friction.
- Collaborated with development, QA, and business teams to align platform engineering priorities with product delivery goals.
- Drove adoption of GitOps and IaC practices within project teams, fostering a culture of automation-first engineering.

EDUCATION

Bachelor of Computer Applications (BCA)

Punjab Technical University, Punjab

CORE COMPETENCIES

Infrastructure Architecture • Platform Engineering • Cloud Cost Optimization • Cross-Team Leadership • Incident Management • Continuous Learning • Problem-Solving • Documentation & Runbooks